Samuel Monteiro Junior

FULL STACK DEVELOPER

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PROFESSIONAL PROFILE

Full Stack Developer with 4+ years of experience building high-performance APIs, microservices and web interfaces in TypeScript, focused on financial products. On the backend I work with Node.js (TypeScript) and Java/Kotlin (Spring Boot), messaging with Kafka and RabbitMQ, PostgreSQL, MongoDB and Redis for caching. On the frontend, React and Vue.js with a strong focus on UX. I deliver end to end: clean, tested code, CI/CD pipelines, containers on Docker and Kubernetes and infrastructure on AWS, with security treated as a requirement from day one. Used to agile teams (Scrum/Kanban) with autonomy and ownership.

TECHNICAL SKILLS

Backend: Node.js (TypeScript, Express, RESTful APIs, WebSockets), Java and Kotlin (Spring Boot), Python (FastAPI), Rust (Tokio, Axum)

Frontend: React, Next.js, Vue.js, TypeScript, Vite, Tailwind CSS, Zustand, TanStack Query, Zod, Shadcn UI, responsive design and UX/UI

Databases: PostgreSQL, MongoDB, Redis (caching, Pub/Sub), TimescaleDB, SQL Server, SQLite, data modeling and query optimization

Architecture & Messaging: Microservices, RESTful APIs, event-driven architecture, Apache Kafka, RabbitMQ, BullMQ, queues and asynchronous processing

Cloud & DevOps: AWS (EC2, S3, RDS), Docker, Docker Compose, Kubernetes, CI/CD (GitHub Actions), Linux, Caddy, Cloudflare Workers, Tailscale, WireGuard

Quality & Security: automated testing (unit, integration and e2e with Jest, Vitest, JUnit, Pytest), Clean Code, SOLID, code review, OWASP, JWT authentication, mTLS, hardening, rate limiting

Methodologies & Languages: Scrum, Kanban, Git Flow • Portuguese (native), English (intermediate)

PROFESSIONAL EXPERIENCE

Full Stack Developer & Automation Engineer · Freelancer April 2022 to Present

- Architected and deployed microservices ecosystems in Node.js (TypeScript) and FastAPI, containerized with Docker, reducing request processing times by 40%.
- Built RESTful APIs and event-driven services with Kafka and RabbitMQ, using Redis as a caching layer and PostgreSQL/MongoDB for persistence.
- Built responsive, user-centered interfaces in React and Vue.js with TypeScript, consuming typed APIs validated with Zod.
- Set up CI/CD pipelines with automated tests and deployments to AWS using Docker and Kubernetes; on Parana-Leads, automated deployment with Caddy and automatic SSL cut new environment setup time by 70%.
- Designed a private mesh network (Tailscale) with mTLS authentication connecting local hosts to cloud servers, securing sensitive data traffic.
- Developed Cortex-Nexus, an automation pipeline (n8n, Python, AI) for lead extraction and qualification with 99.7% deliverability and 3,000+ leads per month.

Support Analyst / Database Administrator · CNI Tecnologia e Software June 2024 to December 2024

- Administered Linux/Windows network security policies and servers, reducing security incidents by 25% through connection hardening and access control.
- Developed Bash and Python scripts to automate relational database backups and hardware performance telemetry, reducing manual monitoring effort by 15 hours per week.
- Optimized queries and maintenance routines in SQL Server and PostgreSQL databases, reducing management report load times by 50%.

Support Analyst & Web Developer · Vip Informática February 2017 to January 2021

- Administered operational databases (PostgreSQL/SQL Server), implementing automatic backup routines and reducing downtime to near zero.
- Developed and maintained internal tools (Node.js/PHP) focused on query and load time optimization, accelerating internal API response times by 35%.
- Led the restructuring of IT support processes and technical documentation, resulting in a 30% reduction in repetitive support tickets.

PROJECTS

TessTrade: Algorithmic Trading Platform `Rust / React / Python / TimescaleDB`

Live platform at tesstrade.com combining social features, a quantitative development environment, and backtesting: multithreaded Rust simulation engine, sandboxed Python indicators connected to the engine via FFI/RPC, and a React workspace with Monaco Editor for authoring strategies in the browser.

6ID: License Ecosystem for Trading Platforms `FastAPI / Next.js / C# WPF / PostgreSQL`

Distributed ecosystem for license management and user authentication on trading platforms: asynchronous FastAPI backend (SQLAlchemy, Asyncpg), Next.js administration console with validation mirrored between Pydantic and Zod, and a Windows WPF client on .NET 8, compiled with ReadyToRun into a single binary.

Sentinel: Edge Bot Firewall `Cloudflare Workers / Durable Objects / JavaScript`

Edge bot firewall running on Cloudflare Workers and Durable Objects: each request is scored by TLS, HTTP version, and browser header signatures, a per-client adaptive rate limit, and a verdict right at the edge (allow, challenge, or block), with logging off the response path and within the free tier.

Aegis: Reverse Proxy and Rate Limiter `Rust / Tokio / Axum / Docker`

Reverse proxy in Rust with per-IP token-bucket rate limiting, timeouts at every stage (including slowloris protection), SIGHUP configuration reload without dropping connections, Prometheus metrics on a separate admin plane, and a FROM scratch Docker image running as a non-root user.

RealScan: External Security Auditor `Cloudflare Workers / Serverless`

Serverless scanner that takes a domain and returns an A to F graded report on HTTPS, security headers, email DNS, subdomains that expose the origin and secrets leaked in client-side code, with an SSRF guard.

EDUCATION

Associate's Degree in Systems Analysis and Development 2024, ongoing

PUCPR (Distance Learning)

Bachelor of Laws (LL.B.) 2016 to 2021

UNISEPE

CC50: Introduction to Computer Science 2023

Harvard University / Fundação Estudar